

Project Specification: Patient Case-Taking Software (MediKiosk)

Problem Statement ID: 26047

Problem Statement Title: Patient Case-Taking Software

Organization: Ministry of Ayush

Department: All India Institute of Ayurveda

Category: Software

Theme: MedTech / BioTech / HealthTech

1. Background & Context

1.1 The Clinical History-Taking Bottleneck in Indian Hospitals

History taking is the structured process of gathering a patient's complaints, present illness history, past medical/surgical history, drug/allergy history, family/personal history, and reviewing their body systems. It is the most vital diagnostic activity in medicine. Classical medical teaching states that a well-conducted history yields the correct diagnosis in **70–80% of cases**, even before physical exams or lab tests.

However, in India's highly burdened public hospital OPDs, time for this critical interaction has collapsed to unsustainable levels. Tertiary government hospitals routinely register **4,000–10,000 OPD patients per day**. Consequently, doctor-to-patient consultation time is frequently reported between **2 and 5 minutes** (a 2017 *BMJ Open* study across 67 countries placed India's average primary-care consultation at just over 2 minutes). Within this tiny window, physicians must elicit history, examine, review old records, diagnose, counsel, and prescribe. This results in under-elicited histories, missed comorbidities, repetitive questioning, and diagnostic errors.

The AYUSH Complexity: Ayurvedic history taking requires a far more extensive framework than allopathic (modern) medicine. Capturing the required depth of data (Trividha, Ashtavidha, Dashavidha Pariksha, etc.) manually within standard OPD time constraints is effectively impossible. This forces practitioners to abbreviate the assessments that define personalized Ayurvedic care.

1.2 The Documentation and Records Fragmentation Problem

Patients in India typically carry physical paper prescriptions, lab reports, discharge summaries, and imaging films from multiple prior providers. During a consultation, doctors must manually sift through these unstructured documents—which are often handwritten, in varying languages, and out of chronological order. This consumes a significant fraction of the already scarce consultation time.

While the **Ayushman Bharat Digital Mission (ABDM)** has established national digital health infrastructure (ABHA IDs, Health Information Exchange, FHIR standards), the "first-mile" problem remains: there is no efficient, patient-facing software platform to capture structured history and digitize documents into the ABDM ecosystem *before* the clinical encounter begins.

1.3 The Opportunity: AI-Powered Digital Clinical Intake Platform

Self-service kiosks have transformed high-throughput industries (bank ATMs, aviation self-check-in, fast-food ordering) by offloading data entry to the user. While patient check-in kiosks exist in developed countries, they are limited to administrative tasks. None perform deep, AI-driven, multimodal clinical history acquisition combined with document digitization.

The convergence of mature technologies now makes this feasible:

- Robust **ASR** for Indian languages/accents (Bhashini / AI4Bharat).
- **LLMs** for conversational clinical history elicitation.
- High-accuracy **OCR** for medical documents.
- **ABDM's FHIR** interoperability.

2. Description of the Problem

2.1 The Problem in Precise Terms

There is no purpose-built, patient-facing software platform that enables patients to independently and comprehensively record their medical history (via natural spoken conversation and guided touch) and simultaneously digitize their existing physical medical documents. The system must generate a structured, physician-ready summary that integrates with the HIS and ABDM ecosystem *before* the patient enters the consultation room.

2.2 Why Existing Solutions Fall Short

- **Hospital Registration Systems:** Capture only demographic data (name, age, token number). They do not elicit clinical history or process medical documents.
- **Mobile Health Apps / Tele-Triage:** Require smartphone literacy, stable internet, and prior enrollment, effectively excluding elderly, rural, low-literacy, and first-visit patients.
- **Manual Nurse-Led Triage:** Human-resource-limited, unable to scale to 5,000+ daily patients, and simply shifts the bottleneck without eliminating it.
- **Generic Document Scanners:** Digitize images but cannot extract, structure, or

chronologically organize clinical content, nor link it to an ABHA record.

2.3 Specific Challenges a Solution Must Overcome

1. **Multilingual/Multi-Accent Voice Capture:** Must work in noisy hospital environments across Hindi, English, and major regional languages for varying literacy levels.
2. **Total Accessibility:** Must be usable by a first-time, non-tech-savvy patient with zero training, utilizing intuitive icon-driven UIs, audio prompts, and conversational guidance.
3. **Accurate Clinical Structuring:** Must convert free-form narration into a standardized, physician-readable history (Chief Complaint, HPI, Past History, etc.) and specific Dashavidha Pariksha parameters for AYUSH settings.
4. **Reliable Document Digitization:** OCR of handwritten/printed prescriptions and reports across multiple languages, with intelligent extraction of diagnoses, meds, and lab values.
5. **Privacy, Consent & Security:** Strict compliance with the **Digital Personal Data Protection (DPDP) Act 2023** and ABDM consent framework to handle sensitive health data securely.

3. Expected Solution: "MediKiosk"

3.1 Solution Overview

"MediKiosk" is the proposed AI-powered clinical history software platform. It allows any patient to record a comprehensive medical history (voice/touch), scan physical documents, and generate a structured, physician-ready summary. This data is pushed to the hospital information system (HIS) and linked to the patient's ABHA record—all completed before the consultation with minimal staff assistance.

3.2 & 3.3 Integrated Software & AI Stack

Module A: Conversational Multimodal History Engine

An AI engine conducting structured clinical interviews through voice and touch.

- **Adaptive Questioning:** Dynamically branches based on answers (e.g., if "chest pain" is stated, it probes using the SOCRATES framework). Mirrors physician reasoning.
- **Dual-Mode Input:** Every question can be answered by speaking or tapping the screen, ensuring usability across all literacy levels.
- **AYUSH History Mode:** An extended interview specifically for Ayurvedic OPDs to capture *Dashavidha Pariksha* and *Ahara-Vihara* assessments.
- **Red-Flag Detection:** The AI flags emergency symptoms (e.g., acute chest pain with dyspnoea) and triggers immediate priority alerts to triage staff, bypassing the routine queue.

Module B: Medical Document Digitization & Intelligence

An integrated scanning pipeline for prior prescriptions, lab reports, and discharge summaries using high-accuracy OCR.

- **Intelligent Extraction:** Pulls diagnoses, prescribed medications (with dosages), investigation results (with reference ranges), and procedural history.
- **Chronological Organization:** Automatically dates and orders documents into a coherent medical timeline.
- **Abnormal-Value Highlighting:** Flags out-of-range lab values and potential drug interactions for the physician.

Module C: Structured History Summary Generator

An AI engine that synthesizes the conversation and documents into a single, concise, standard clinical summary, ready on the doctor's screen when the patient enters.

- **Standard Clinical Format:** Chief complaint → HPI → Past medical/surgical → Drug & allergy → Family → Personal → ROS → Prior investigations.
- **Editable & Verifiable:** The physician retains full control. The summary is a draft to be accepted, amended, or rejected; the AI *never* acts as an autonomous diagnosis.
- **Bilingual Output:** Patient-facing audio confirmation is provided in the local language; the physician-facing summary is in English/Hindi.

Module D: Consent, Privacy & ABDM Integration

A robust security layer compliant with the DPDP Act 2023 and ABDM frameworks via FHIR APIs.

- **Consent-First Design:** Features granular, revocable consent with audio explanations specifically designed for low-literacy patients.
- **Secure Processing & Session Termination:** AI processing is secure, and all temporary session data is permanently cleared immediately after submission.

3.4 End-to-End Patient Journey

1. **Step 1 — Identify:** The patient logs into the platform, enters/scans their ABHA ID or Aadhaar (or registers as new), selects a language, and grants audio-guided consent.
2. **Step 2 — Converse:** The AI conducts an adaptive voice and touch interview to capture the chief complaint and full history. Red flags trigger priority triage.
3. **Step 3 — Scan:** The patient uploads prior physical prescriptions, lab reports, and discharge summaries. The AI digitizes, structures, and timelines them.
4. **Step 4 — Summarize & Route:** The AI generates a structured history summary, links it to ABHA, pushes it to the HIS, and updates the patient's digital record.
5. **Step 5 — Consult:** The physician reviews the complete history on their screen in seconds, edits/confirmes it, and devotes the full consultation time entirely to examination, reasoning, and patient counseling.

Glossary of Terms for General Readers

To ensure this document is easily understood, here is a guide to the technical, medical, and Ayurvedic terms used throughout the text.

Medical & Clinical Terms

- **OPD (Outpatient Department):** The hospital section where patients visit for consultations without staying overnight.
- **Comorbidities:** When a patient has two or more diseases or medical conditions at the same time.
- **HPI (History of Present Illness):** A detailed interview regarding the patient's current symptoms and the progression of their current illness.
- **ROS (Review of Systems):** A systematic checklist of questions to uncover dysfunction in different body systems (e.g., respiratory, cardiovascular).
- **Dyspnoea:** Medical term for shortness of breath or difficulty breathing.
- **Triage:** The process of prioritizing patients based on the severity of their condition.
- **SOCRATES:** A medical checklist used by doctors to evaluate pain (*Site, Onset, Character, Radiation, Associations, Time course, Exacerbating/relieving factors, Severity*).
- **Pathogenesis:** The manner of development of a disease.

Ayurvedic (AYUSH) Terms

- **AYUSH:** Traditional medicine systems in India: Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy.
- **Pariksha (Trividha, Ashtavidha, Dashavidha):** Traditional Ayurvedic clinical examination frameworks (3-fold, 8-fold, and 10-fold patient assessments).
- **Assessment Parameters:**
 - *Prakriti* (natural body constitution)
 - *Vikriti* (current disease or imbalance)
 - *Agni* (digestive capacity)
 - *Koshtha* (nature of the bowel)
 - *Ahara-Vihara* (diet and lifestyle)
 - *Nidana* (causative factors)
 - *Samprapti* (pathogenesis/disease development)
 - *Sara* (tissue quality), *Samhanana* (body build), *Pramana* (body proportions), *Satmya* (adaptability), *Sattva* (mental strength), *Ahara Shakti* (food intake capacity), *Vyayama Shakti* (exercise capacity), *Vaya* (age).

Technological & Government Terms

- **ABDM (Ayushman Bharat Digital Mission):** India's national digital health infrastructure initiative.
- **ABHA ID:** Ayushman Bharat Health Account; a unique digital health ID for Indian citizens.
- **HIS / EMR:** Hospital Information System / Electronic Medical Record software.

- **FHIR (Fast Healthcare Interoperability Resources):** A global data standard ensuring different healthcare computer systems can share data securely.
- **ASR (Automatic Speech Recognition):** AI technology (like India's *Bhashini* or *AI4Bharat*) that converts spoken words into text.
- **OCR (Optical Character Recognition):** Technology that reads physical documents and turns them into digital text.
- **DPDP Act 2023:** Digital Personal Data Protection Act of India, governing data privacy.